

AL-AHSA, Saudi Arabia

WMO: 404200

Lat: 25.285N

Lon: 49.485E

Elev: 588

StdP: 14.39

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.3	46.5	14.3	11.6	58.7	19.2	14.8	64.8	26.1	61.6	23.5	63.5	6.2	330	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.2	116.9	72.4	115.2	72.5	114.3	72.4	83.6	100.7	81.2	99.8	79.0	101.3	10.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.3	160.8	92.7	76.8	142.5	92.7	73.1	125.8	91.6	48.5	100.0	45.7	99.8	43.1	101.1	99.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.1	21.9	19.3	DB	38.8	120.3	3.0	1.5	36.6	121.4	34.9	122.3	33.2	123.1	31.0	124.2	(4)
(5)			WB	32.3	86.5	2.9	4.3	30.2	89.5	28.5	92.0	26.9	94.4	24.8	97.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.3	60.4	64.5	72.2	82.2	92.8	98.2	100.6	99.9	94.1	85.4	72.6	63.3		
	DBStd	15.22	5.46	6.03	6.05	5.78	4.32	2.91	2.51	2.45	3.55	4.13	5.81	5.43		
	HDD50	4	3	0	0	0	0	0	0	0	0	0	0	1		
	HDD65	351	162	74	9	0	0	0	0	0	0	0	9	97		
	CDD50	11785	325	407	688	966	1325	1446	1570	1548	1322	1097	678	413		
	CDD65	6658	20	61	233	516	860	996	1105	1083	872	632	236	44		
	CDH74	105018	224	627	2442	6577	13909	17417	19761	19111	14071	8338	2165	376		
(13)	CDH80	74059	37	196	1116	3880	9730	13110	15297	14649	9912	5143	918	71		
(14)	Wind	WSAvg	7.6	7.5	8.2	8.2	7.8	7.9	9.1	7.2	6.4	5.4	6.5	7.1		
(15) Precipitation	PrecAvg	3.4	0.6	0.4	0.6	0.5	0.1	0.0	0.0	0.0	0.0	0.0	0.6	0.6		
	PrecMax	10.5	2.4	2.9	4.4	1.7	1.2	0.1	0.0	0.7	0.0	0.3	3.8	4.5		
	PrecMin	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.3	0.6	0.6	0.9	0.5	0.3	0.0	0.0	0.1	0.0	0.1	1.1	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.2	91.5	100.2	107.7	115.1	118.5	119.8	118.5	114.8	107.7	96.0	85.8		
		MCWB	59.2	63.6	65.1	70.1	71.2	71.3	72.3	73.4	71.8	68.7	66.4	64.1		
	2%	DB	79.9	85.8	95.1	104.0	112.3	116.3	117.0	116.6	112.5	104.2	92.8	82.0		
		MCWB	58.4	60.5	63.7	67.6	71.1	71.3	72.2	73.7	71.9	68.1	65.5	61.8		
	5%	DB	75.6	81.0	91.1	100.6	109.7	114.4	115.3	115.0	109.7	102.0	89.4	78.4		
		MCWB	57.0	58.6	62.8	67.0	70.0	71.5	72.0	73.7	71.2	68.3	65.4	60.3		
	10%	DB	71.8	77.1	86.1	97.0	107.5	111.9	113.6	113.2	107.8	99.0	85.8	74.9		
		MCWB	55.6	57.5	61.1	65.7	69.5	70.5	71.8	73.4	70.7	67.5	63.7	59.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.4	67.5	70.5	74.5	78.9	81.0	86.1	87.8	83.5	78.7	73.4	68.8		
		MCDB	73.3	79.3	87.5	98.1	106.5	106.7	103.5	103.0	96.1	88.9	81.8	74.8		
	2%	WB	62.7	64.7	67.9	72.0	75.6	77.9	81.9	84.7	80.7	76.1	70.6	65.9		
		MCDB	70.8	74.9	83.7	94.3	103.8	107.8	101.2	102.2	94.6	87.8	80.7	74.0		
	5%	WB	60.6	62.5	65.8	70.1	73.8	75.5	78.9	82.5	78.5	74.0	68.7	63.8		
		MCDB	69.4	74.7	80.5	91.1	102.2	106.6	105.1	99.3	95.7	88.0	80.6	73.3		
	10%	WB	58.1	60.4	63.8	68.3	71.8	73.2	76.0	80.0	76.1	71.9	66.9	61.6		
		MCDB	68.9	72.7	79.3	88.8	100.1	105.6	105.4	101.4	95.8	88.2	79.7	71.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.5	23.0	24.4	26.0	27.6	28.0	27.2	28.0	28.1	27.2	22.4	21.0		
		MCDBR	27.2	29.0	31.2	31.2	30.7	30.7	29.0	29.5	29.4	29.5	26.8	25.8		
		MCWBR	12.7	12.8	13.2	11.8	11.5	11.5	10.9	10.9	11.7	11.9	12.0	12.6		
	5% WB	MCDBR	20.5	22.9	24.3	26.5	28.0	29.0	28.0	28.0	28.1	26.4	22.2	20.9		
		MCWBR	11.2	11.9	12.0	11.6	12.8	13.0	13.3	13.0	13.7	12.6	11.4	11.4		
(40) Clear-Sky Solar Irradiance	taub		0.518	0.576	0.643	0.704	0.712	0.699	0.707	0.658	0.580	0.519	0.526	0.489		
	taud		1.714	1.598	1.478	1.415	1.396	1.399	1.422	1.506	1.622	1.738	1.727	1.784		
	Ebn at Noon		225	223	217	209	207	209	207	216	229	235	221	227		
	Edh at Noon		66	80	95	103	105	103	101	93	81	68	65	59		
(44) All-Sky Solar Radiation	RadAvg		1317	1566	1824	2009	2262	2432	2305	2227	2049	1750	1355	1244		
	RadStd		79	87	112	106	67	80	59	38	37	52	84	81		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.74	N/A	-3.69	+1.03	N/A	N/A	N/A	N/A	+171	+158	(46)
(47)	Regional (4 neighbors)	+0.85	N/A	-3.29	+1.35	N/A	N/A	N/A	N/A	+306	+283	(47)

Nomenclature: See separate page